

# Minseong Cho

✉ mscho713@ajou.ac.kr  
☎ +82 10-8803-7786  
🏢 Softlab, Ajou University  
📍 Suwon, South Korea  
📁 Portfolio  
🎓 Google Scholar



## Research Interests

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Active matter dynamics · Elastic Leidenfrost effect · Mechanics of Moisture-Driven Hair Frizz

## Personal Statement

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I am an M.S. student in Mechanical Engineering and a graduate researcher at Ajou University. My research focuses on understanding how soft and deformable systems move, adapt, and self-organize under nonequilibrium conditions, driven by energy input, environmental changes, and interactions with their surroundings.

Since my undergraduate years, I have developed a hands-on research approach by designing experiments, building quantitative image-analysis methods, and interpreting complex behaviors through mechanics and physics. I enjoy starting from subtle experimental observations, developing the analytical tools needed to characterize them, and uncovering the physical principles that govern their behavior.

As a researcher, I am particularly drawn to unfamiliar problems that require patience, technical flexibility, and careful interpretation. Rather than being defined by a single material or research topic, I aim to develop a versatile way of thinking that integrates experiments, quantitative analysis, and physical reasoning to reveal the physics behind complex phenomena across a wide range of systems.

## Education

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- **M.S. in Mechanical Engineering** Sep 2025 – Present  
Ajou University, Suwon, South Korea
  - **Advisor:** Prof. Jonghyun Ha
- **B.S. in Mechanical Engineering** Mar 2020 – Aug 2025  
Ajou University, Suwon, South Korea

## Research Experience

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- **Graduate Research Student – Softlab, Ajou University** Sep 2025 – Present
  - **Mechanics of Moisture-Driven Hair Frizz**  
**Focus:** Moisture-driven changes in strand morphology.  
**Techniques:** Centerline-based geometric analysis and simulation-assisted interpretation.  
**Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung
  - **Boundary-Driven Active Matter in Soft Hydrogels**  
**Focus:** Leidenfrost-induced dynamics in soft hydrogel systems.  
**Techniques:** Image-based tracking, curvature analysis, and visualization scripts.  
**Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung
- **Undergraduate Research Student – Softlab, Ajou University** Mar 2023 – Aug 2025
  - **Boundary-Driven Active Matter in Soft Hydrogels**  
**Focus:** Leidenfrost-related soft-matter dynamics.  
**Techniques:** Image-based tracking, quantitative analysis, and figure preparation.  
**Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung

## Conferences and Presentations

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- **APS Division of Fluid Dynamics Meeting (Houston, TX, USA)** Nov 2025
  - Leidenfrost-induced active matter dynamics of spherical hydrogels
- **Asian Congress of Fluid Mechanics (Seoul, South Korea)** Sep 2025
  - Leidenfrost-induced active matter dynamics of spherical hydrogels
- **KSME Annual Conference (Jeju, South Korea)** Nov 2024
  - Explosive motion of spherical hydrogel via the Leidenfrost effect

## Awards & Honors

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- **Dean's List** – Ajou University Spring 2024, Spring 2025  
Recognized for outstanding academic performance (top 5% of the department).
- **Capstone Design Competition** – Ajou University June 2025  
2nd Place · KRW 1,000,000 prize  
**Project:** Compliant-mechanism-based knee assistive device for degenerative osteoarthritis prevention.
- **Gyeonggi-do Assistive Device Idea Competition** Nov 2025  
3rd Place · KRW 300,000 prize  
**Organized by:** Gyeonggi Province Rehabilitation Engineering Service Support Center (South Korea)  
**Project:** Compliant-mechanism-based knee assistive device for older adults.
- **APS DFD Travel/Enabling Grant** 2025  
US\$750 travel cost reimbursement  
**Organized by:** APS Division of Fluid Dynamics, American Physical Society  
**Purpose:** Travel support for participation in the APS Division of Fluid Dynamics Meeting.
- **Gallery of Fluid Motion** Apr 2026  
Haecheon Choi GFM Award · KRW 400,000 prize  
**Organized by:** The Korean Society of Mechanical Engineers, Fluid Engineering Division  
**Title:** Explosive motion of spherical hydrogel via the Leidenfrost effect  
**Authors:** Minseong Cho, Yeonsu Jung, Jonghyun Ha

## Publications

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- **Boundary-driven active matter and stochastic rectification via Leidenfrost impulses**  
Minseong Cho, Yeonsu Jung, and Jonghyun Ha  
☒ In preparation
- **Poroelectric swelling time as a quantitative marker of hair damage and treatment**  
Jaehyuk Cho, Minseong Cho, Selin Cho, Haesoo Park, Hyun-Na Kim, and Jonghyun Ha  
☒ In preparation

## References

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### Prof. Jonghyun Ha

Assistant Professor, Department of Mechanical Engineering, Ajou University, Suwon, South Korea

E-mail: [hajh@ajou.ac.kr](mailto:hajh@ajou.ac.kr)

Scholar Profiles: [Personal Page](#) | [Google Scholar](#)

### Dr. Yeonsu Jung

Postdoctoral Research Fellow, School of Engineering and Applied Sciences, Harvard University, Cambridge, MA, USA

E-mail: [jung@seas.harvard.edu](mailto:jung@seas.harvard.edu)

Scholar Profiles: [Personal Page](#) | [Google Scholar](#)